Module3

- Moral Framework of Justice in AI: on the Limits, Failing and Ethics of Fairness, Accountability in Computer Systems-Responsibility and AI, The concept of Handoff as a Model for Ethical Analysis and Design.

- AI systems are increasingly used in important areas such as hiring, loan approval, criminal justice, healthcare, and education.
- Because these systems influence people's rights and opportunities, questions of justice and fairness become very important.
- The moral framework of justice in AI examines whether AI systems treat individuals and groups ethically, fairly, and responsibly.

Justice in AI:

- Justice generally refers to fairness, equality, and protection of human rights.
- In the context of AI, justice means:
  - AI systems should not discriminate.
  - Individuals should receive fair treatment.
  - Human dignity and rights must be respected.
  - Decisions should be accountable and explainable.
- AI systems do not have moral values of their own.
- They reflect the values and assumptions of the people who design and deploy them.

- **Understanding Fairness in AI**
  - Fairness in AI usually refers to the absence of bias and discrimination.
- An AI system is considered fair when:
    - It does not favor or disadvantage individuals based on race, gender, religion, caste, or economic background.
    - It provides equal opportunity.
    - It applies consistent standards in decision-making.
- However, fairness is complex because it can be defined in different ways, and these definitions may sometimes conflict.

- **Defining Fairness**
- **Individual Fairness**
  – Similar individuals should be treated similarly.
- **Group Fairness**
  – Equal outcomes across different demographic groups.
- **Procedural Fairness**
  – Transparent and consistent decision-making processes.
- **Distributive Justice**
  – Equitable allocation of benefits and burdens.

Biased Training Data in Healthcare AI

- **Ethical Implications of Biased Training Data**
- **Sources of Bias**
  - Historical data reflects past inequities.
  - Under-representation of minority groups.
  - Socioeconomic and geographic data gaps.
  - Label bias from medical professionals.
- **Ethical Consequences**
  - Higher misdiagnosis rates in marginalized groups.
  - Perpetuates systemic health disparities.
  - Erodes patient trust in AI systems.
  - Unequal access to accurate AI-assisted care.

Healthcare AI Bias – Real World Impact

- **Pulse Oximetry Devices**
  - Devices designed and tested primarily on lighter skin tones showed less accurate readings for darker-skinned patients.
  - This contributed to delayed COVID-19 diagnoses and treatment.
- **Dermatology AI Algorithms**
  - Skin cancer detection models trained mostly on light-skinned datasets performed worse on darker skin.
  - Resulted in higher rates of missed malignant lesions.
- **Kidney Function Estimation**
  - Race-based adjustments in eGFR algorithms historically classified Black patients as having better kidney function than they did.
  - This delayed referrals for specialist care and transplant eligibility.
- **Key Takeaway:**
  Bias in healthcare AI is not just a technical flaw — it is an ethical failure that can reinforce systemic inequality and cause real harm.

Limits & Failings of AI Fairness

- **The Impossibility Theorem**
  - Mathematical proof shows that multiple fairness criteria (e.g., equal accuracy, equal false positive rates, calibration) cannot all be satisfied at the same time.
  - Trade-offs are unavoidable.
- **Fairness is Context-Dependent**
  - What counts as fair varies across cultures, legal systems, and use cases.
  - No single universal fairness metric exists.
- **Proxy Discrimination**
  - Removing protected attributes (e.g., race, gender) does not eliminate bias.
  - Correlated proxies (e.g., zip code, name) can still produce discriminatory outcomes.
- **Feedback Loops**
  - Biased AI decisions create biased outcomes.
  - These outcomes reinforce biased training data, compounding inequality over time.
- **Measurement Problems**
  - Training data often reflects existing societal biases.
  - Fairness evaluations may encode these biases instead of correcting them.

Human Oversight in AI Systems

- **Role of Human Oversight in Maintaining Fairness & Accountability**
- **Algorithmic Auditing**
  - Human experts periodically review AI decisions for disparate impact.
  - Ensures no group is systematically disadvantaged.
- **Interpretability & Explainability**
  - AI systems should provide human-understandable reasoning.
  - Supports accountability and meaningful review.
- **Diverse Review Boards**
  - Include stakeholders from affected communities.
  - Bring lived experience into technical governance.
- **Override Mechanisms**
  - Clear human override protocols for critical decisions.
  - Essential in high-stakes areas like justice and healthcare.
  - Ensures critical decisions remain under human control.

- Different moral approaches:
- **Equality-Based Ethics**
  - Treat everyone the same.
- **Equity-Based Ethics**
  - Treat people differently to correct disadvantages.
- **Rights-Based Ethics**
  - Protect privacy and human dignity.
- **Utilitarian Ethics**
  - Maximize overall social benefit.
- “Different ethical theories lead to different AI design choices.”

Accountability in Computer Systems.

- Computer systems are widely used in modern society for important decisions such as:
  - Banking transactions
  - Medical diagnosis
  - Autonomous vehicles
  - Recruitment systems
  - Criminal justice
- When these systems make mistakes or cause harm, an important question arises:
  - **Who is responsible for the decision made by the system?**
    - This leads to the concept of **Accountability in Computer Systems**.

- Accountability ensures that there is **clear responsibility for the design, operation, and outcomes of computer systems**, especially those using Artificial Intelligence.

- Accountability helps ensure that technology is used **safely, ethically, and responsibly.**

- **Definition of Accountability**
    - **Accountability** refers to the ability to **identify, trace, and hold responsible individuals or organizations for actions taken by computer systems**.
- In simple terms:
    - **Accountability means ensuring that someone is responsible when a computer system causes an error, harm, or unfair decision.**
- Accountability helps in:
    - building trust in technology
    - ensuring ethical use of systems
    - preventing misuse of AI technologies

Importance of Accountability in AI

- Accountability is very important in artificial intelligence systems for several reasons.
    - First, it helps **build trust** between users and technology. People are more likely to trust AI systems when they know that the system's decisions can be explained and verified.
    - Second, accountability helps **prevent misuse of technology**. Without accountability, AI systems could be used in ways that harm individuals or society.
    - Third, accountability helps **identify and correct errors**. When systems fail or produce incorrect results, accountability allows developers to understand what went wrong and improve the system.
    - Finally, accountability helps ensure that organizations follow **ethical and legal standards** while developing AI technologies.

Components of Accountability:

- There are several important principles that support accountability in AI systems.
- **Transparency**
  - Transparency means that the system's operations and decision-making processes should be clear and understandable.
    - Users should know **how data is collected, processed, and used by the system.**
- **Explainability**
  - Explainability means that AI systems should provide **clear explanations for their decisions.**
    - For example, if an AI system rejects a loan application, it should explain the reasons for the rejection.

- **Fairness**
  - AI systems should treat all individuals equally and should not produce biased or discriminatory results.
    - Bias can occur if the training data used to develop AI models contains historical inequalities.
- **Responsibility**
  - Developers and organizations must take responsibility for the outcomes produced by their AI systems.
    - They should ensure that systems are properly tested and monitored.
- **Auditability**
  - AI systems should allow independent auditing and evaluation to ensure that they operate correctly and ethically.

Challenges in AI Accountability:

- Although accountability is important, implementing it in AI systems is not always easy.
  - One major challenge is the use of **complex machine learning models**, especially deep learning systems.
  - These models often function as **black boxes**, meaning it is difficult to understand how they arrive at a particular decision.
  - Another challenge is **data bias**. AI systems learn from data, and if the data contains bias, the system may produce unfair results.
  - There is also the challenge of **multiple stakeholders** involved in AI development, including developers, data scientists, companies, and users.
  - Determining who is responsible when something goes wrong can be complicated.
  - Additionally, **laws and regulations related to AI are still evolving**, making accountability more difficult to enforce.

Ensuring Accountability in AI Systems

To ensure accountability in AI systems, organizations should follow several practices.

- They should develop **ethical AI guidelines** and ensure that AI systems are designed responsibly.
- Regular **algorithm audits** should be conducted to detect bias and errors.
- Proper **documentation of data and models** should be maintained to improve transparency.
- **Human oversight** should be included in critical decision-making processes.
- Finally, governments and regulatory bodies should establish **clear policies and standards for AI development.**

- **Role of Engineers and Developers**
  - As future engineers and AI professionals, have an important role to play in building responsible technology.
  - They should focus on developing systems that are **fair, transparent, and explainable.**
  - carefully select datasets, test models thoroughly, and consider the **social and ethical impact** of our technologies.
  - Responsible AI development helps create systems that benefit society while minimizing risks.

Examples of Accountability Issues in AI

- **Example 1: Self-Driving Cars**
- Companies such as **Tesla** and **Waymo** are developing autonomous vehicles.
  - If a self-driving car causes an accident, responsibility could lie with:
    - the car manufacturer
    - the software developer
    - the owner of the car
    - the AI system itself
    - This creates legal and ethical challenges.

Example 2: AI in Healthcare

- AI systems are used for:
  - disease prediction
  - medical image analysis
  - treatment recommendations
- If an AI system suggests an incorrect diagnosis, responsibility could lie with:
    - the hospital
    - the doctor using the system
    - the AI developers
  - Currently, doctors are expected to **verify AI recommendations** before making final decisions.

Methods to Ensure Accountability in AI Systems

- Several strategies can improve accountability.

1. Explainable AI (XAI)

  - Explainable AI focuses on making AI systems easier to understand.
  - These systems provide reasons for their decisions.

- Example:
  Explaining why a loan was rejected.

2. AI Governance Policies

  - Organizations develop internal guidelines for responsible AI use.
  - Companies like **Google** and **IBM** have published AI ethics principles.

3.Documentation

- Developers should document:
  - datasets used
  - algorithms applied
  - system limitations
- This helps identify problems when they occur.

4.Regular Audits

  - Independent experts should periodically examine AI systems to check for:
    - bias
    - accuracy
    - fairness

5.Legal Regulations

  - Governments are introducing laws to regulate AI technologies.
- These laws require organizations to take responsibility for the systems they deploy.

The Concept of Handoff as a Model for Ethical Analysis and Design

- Modern computer systems are **complex and involve many participants** such as:
    - software developers
    - system designers
    - organizations
    - users
    - regulators
- During the development and use of technology, **responsibility is often transferred from one person or group to another**.
- This transfer of responsibility is called **Handoff**.

- **A handoff** occurs when:
  - A human gives control to an AI system
  - An AI system returns control to a human
  - Control shifts during task execution
- It focuses on **“who is responsible at each stage of a task.”**
- **Handoff as an Ethical Model**
- The handoff model helps evaluate:
  - **Who makes decisions?**
  - **Who is accountable for outcomes?**
  - **Whether the transition of control is safe and clear**

- **Types of Handoffs**
- **1. Human → AI**
- Example: Activating autopilot in a vehicle
- Issue: Over-reliance on AI
- **2. AI → Human**
- Example: AI requests human intervention
- Issue: Human may not be attentive or ready
- **3. Shared Control**
- Both human and AI operate together
- Issue: Confusion about authority and responsibility

- **Ethical Issues in Handoff**
- **1. Responsibility Gap**
  - Unclear who is responsible for failure
- **2. Automation Bias**
  - Humans blindly trust AI decisions
- **3. Lack of Awareness**
  - Users may not know when control changes
- **4. Timing Problems**
  - Delayed or premature handoffs can cause harm

- **Design Principles for Ethical Handoffs**
- To ensure ethical AI systems, handoffs must be designed carefully:
- **Clarity:** Clearly indicate who is in control
- **Transparency:** Explain why the handoff occurs
- **Smooth Transition:** Provide warnings before switching
- **Human Readiness:** Ensure users can take control
- **Accountability:** Clearly assign responsibility

- **Why is Handoff Important?**
- “In real-world systems, problems happen because:
- People don’t clearly understand their responsibility
- Information is not properly shared
- One group blames another

- The handoff model helps us:
- Understand responsibility clearly
- Avoid confusion
- Design better and safer systems”

- The **Handoff Model** helps us analyze **ethical responsibilities in technological systems**.
- It helps answer questions like:
  - Who is responsible for a system’s design?
  - Who is responsible when something goes wrong?
- Instead, responsibility **moves between different actors at different stages**.
- Example stages:
  - Design stage
  - Development stage
  - Deployment stage
  - Usage stage
  - Maintenance stage
- At each stage, **responsibility is handed off**.

- **Stages of Handoff**
- “Let us look at how responsibility moves in different stages.”
- **1. Design Stage**
- “In this stage, designers decide what the system should do.
- They must think about:
  - Is the system fair?
  - Will it cause harm?
- If design is wrong, the whole system can fail.”

- **2. Development Stage**
- “Here, developers write code and build the system.
- They must:
  - Write correct code
  - Avoid bugs and errors
  - If they make mistakes, the system may not work properly.”

- **3. Deployment Stage**
- “In this stage, companies release the system.
- They must:
  - Test the system
  - Make sure it is safe
- If they release it too early, it can cause harm.”

- **4. Usage Stage**
- “Now users use the system.
- Users must:
  - Use it properly
  - Not misuse it
- But users should also understand that the system is not always perfect.”

- “After release, the system needs updates.
- Engineers must:
  - Fix errors
  - Improve the system
  - Ignoring problems can make things worse.”

- **Ethical Problems at Handoff Points**
- “Most ethical issues happen at the *handoff points*.
- Some common problems:
  - No one takes responsibility
  - Poor communication
  - Users trust systems too much
  - Loss of human control
- These problems happen when responsibility is not clearly passed.”

- **Simple Example**
- “Let’s take an example of an AI system:
- Data is collected
- Model is built
- Company deploys it
- Users depend on it
- If the system gives wrong results, we cannot blame just one person.
- The problem may have happened at any stage.”

- Summary :
- Handoff means transfer of responsibility
- Responsibility changes at each stage
- Ethical problems occur when handoffs are not managed properly
- So as future engineers, always remember: Good systems are built not just with good code, but with **clear responsibility at every stage**.”